

Exploring the design space for Crowdsourcing Journey eXperience

Selin Zileli
s.zileli@soton.ac.uk
University of Southampton
Southampton, United Kingdom

Richard Gomer
r.gomer@soton.ac.uk
University of Southampton
Southampton, United Kingdom

m.c. schraefel
mc@ecs.soton.ac.uk
University of Southampton
Southampton, United Kingdom

ABSTRACT

This paper aims to establish a design space for tools to crowdsource data about Journey eXperience (JX). Understanding JX could provide a range of benefits in various domains, including transport planning, urban design, and individual route planning, and support efforts to achieve more sustainable transport systems. A key challenge in studying JX is finding effective ways to collect data. We present initial results from a diary study that explores what JX means and how JX data might be collected. We show how the experience of journeying through space differs from experiences of being in space, provide an initial description of factors that influence JX, and offer high-level insights of relevance to the design space of JX data collection.

CCS CONCEPTS

• **Human-centered computing** → **User centered design**.

KEYWORDS

user experience, diary, journey, behaviour, senses, emotions, sustainability, transport

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1 INTRODUCTION

Encouraging the use of sustainable mobility is an important part of transitioning societies to net-zero carbon emissions and limiting global heating [28]. Sustainable transport forms an important part of the United Nations' Sustainable Development Goals [2].

Improving the accessibility and appeal of sustainable transportation options is crucial for increasing their usage and achieving an emission-free transportation system [31]. Two approaches to doing so are apparent in previous work. The first is through improvements in the built environment or transport infrastructure [14], for example, providing better bus routes, building new cycle paths, or making existing routes more pleasant. The second is in individual

journey planning, combining information about travel needs and mobility availability (like bus routes or the footpath network) to create individualised plans for journeys, such as through tools like Google Maps or CityMapper [26]. Both applications involve predicting how transport infrastructure can meet the needs of one or more actual or anticipated transport users and drawing on various data sources to model actual or possible mobility provisions.

Digitalisation provides opportunities for more sustainable urban mobility and is associated with three specific trends: mobility as a service [9], electrification and automation [8]. According to Creutzig et al. [8], digitalisation techniques (such as open data, big data, sensor technologies, crowdsourcing and so on) can enable a shift to decarbonisation in cities when combined with design strategies for walking, cycling, and efficient public transport.

In this research, we are interested in developing tools that could support strategic planning and individual route planning that improve what might be termed "Journey eXperience" (JX), *"the experience that is shaped through the interaction between travellers and their mode of transport, other objects and technologies ... and other travellers"* [18]. We hypothesise that better understanding passengers' JX would allow for more targeted interventions in transport systems and urban planning, as well as improvements to individual journey planning tools to help individuals to navigate sustainable transport networks in more convenient or pleasant ways, and would hence support efforts to encourage modal shift to sustainable transport.

Specifically, given the increasing role of data within both strategic approaches (e.g. [25]), we are interested in building upon prior work on crowdsourcing to make datasets about JX more readily available. Collecting JX data about (often complex) daily journeys is challenging, logistically and conceptually. Current approaches such as travel surveys are costly and time consuming, but provide a time- and granularity-limited snapshot [1]. Prior work on crowdsourcing mobility data has looked at the potential for re-purposing data from location-based social networks (LBSNs) [29, 33] or of using sensed data collected by citizens using their smartphones to create relevant datasets [12, 17].

However, we observe that the vast majority of datasets used in previous work provide data about locations, rather than journeys. While a naive view of a journey might treat a journey as presence in a series of locations, prior work on JX (e.g. [18]) frames JX as shaped by the interactions between traveller, mode of travel, the purpose of travel and other objects, as well as surroundings (i.e. the city itself), and we, therefore, hypothesise that elements of JX cannot be predicted just through data about space and place, but must be viewed through the lens of the journey itself.

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Some prior work does consider journeys, specifically, and Ibáñez et al. [17] present a hybrid human-AI system for crowdsourcing "modal split" data (data about which transport modes are being used for journeys) from citizens as an alternative to telephone-based surveys. Their approach combines automated classification of smartphone data with travellers' reviews and corrections. We hypothesise that including humans in the loop could also allow collecting data about human experiences, and serve as a means to verify and correct more objective journey data.

In this paper, to that end, we present an overview of a diary study in which we sought to begin understanding how we might design more formal crowdsourcing tools for collecting JX data. Our aim is to understand to what extent the experience of journeying through spaces might differ from the experience of being in them, to identify factors that seem to affect JX, and to understand more about the conceptual challenges in designing for JX data collection.

2 BACKGROUND & RELATED WORK

"Journey Experience" as a term is used by Jain & Clayton [18] in their review of journey experience as it pertains to tasks that can be undertaken during transportation, such as work and leisure, focussed primarily on journeys taken on powered transport. Middleton [23] explores the "socialities of everyday urban walking," providing an important account of how the social context around a pedestrian journey impacts pedestrians' experiences.

Technology interventions that collect data about journeys, or (more frequently) about the environments in which journeys take place, are already abundant and come in several different forms. Civic technologies such as fixmystreet [6, 22, 30] and pothole reporters [4], for example, are tools that collect user-generated data on spatial features and local services; although not framed directly as JX data collection tools, their focus on civic spaces in which journeys take place and on civic infrastructure which facilitates those journeys, means that they are closely related. Such platforms enable citizens to report a variety of neighbourhood irregularities [6], ranging from broken traffic lights to abandoned vehicles. However, citizens' involvement with these systems is restricted to addressing environmental "defects", without encouraging a more profound re-conceptualisation of what the environment could be.

Tools for planning transport infrastructure, and tools for individual journey planning, typically use distance and/or time-based utility functions for comparing possible journeys, preferring temporally or geographically shorter routes over longer ones. There has been some interest in altering those utility functions to include route attractiveness or safety. Chen et al. [5] define an algorithm for finding routes that considers time-varying utility. They use the example of Victoria Harbour in Hong Kong, which they propose would be more scenic (and hence have a higher utility) at night, but do not propose a method for collecting that utility information. Folco et al. [14] propose a transport network development model that can be tuned to give different weights to demand-satisfaction and safety, demonstrating how different data sets (representing different priorities) may be used to inform the development of mobility networks.

Attempts to use alternative utility functions have constrained dataset availability. For example, Herzog et al. [16] were able to use

data about trees to predict route attractiveness but found that data on air pollution was not available at a suitable granularity. They note that *"a big challenge for the developers of smart tourism and mobility applications is the availability of such [route attractiveness] data."*

Lim et al. [19], also in the context of tourism, point to the importance of data collection and dataset availability for tour recommendations and survey the role of location-based social media in providing useful data, but focus primarily on understanding points of interest (POIs) rather than the experience of journeying between them.

Another crowdsourcing example that looks into a level of emotional experience by Quercia et al. [27] mapped pleasant streets in London. Instead of users providing photos and reporting problems, they used a crowdsourcing platform where contributors rated photos of street views for pleasantness based on three measures: sonic qualities, visual appearance and happiness. Based on these ratings, the authors created a route recommendation example and tested it with participants who knew the area to see whether they were accurate. The research brings together spatial and experiential aspects of journeys. However, the authors note that their approach lacks situational factors such as temporal dynamics, weather conditions and time of the day, which can considerably affect travellers' experiences.

Ibáñez et al. [17] use a combination of crowdsourcing and machine learning to create "modal split" data - data that shows which transport modes are being used to make trips within a city. Although related to sustainable transport, which motivates our own research, their focus is specifically on the mode of transport used to make a journey, rather than on the more subjective experience of the travellers.

While many digital technologies involve citizens during data collection, the data provided by contributors is not necessarily connected with their experience of the space or their journey. Individuals' perceptions of the environment, as Ewing and Handy [13] indicate, have a complex relationship with physical aspects, and collecting data solely on these physical features may not tell much about individuals' perceptions of urban space. In the case of urban mobility, the connection of intangible qualities and physical features are important and hence Vallejo-Borda et al. [34] suggest that future studies should involve emotional states, participants' responses and sensory experiences.

3 METHODOLOGY

We conducted a structured diary study to learn more about people's journey experiences over 14 days. Structured diary methodology involves asking participants to keep a journal of their experiences, behaviours, and thoughts over a specific period of time [10].

We created a structured diary kit, which includes prompts to guide participants in reflecting on their daily journey experiences through textual and non-textual responses, including maps, stickers, and their smartphone camera. The prompts in the diary aim to elicit specific information and responses from participants; they range from open-ended free-text questions to fill-in-the-blank exercises, map marking, taking photos, applying stickers, drawing,

and collage. These prompts were inspired by a human-centred design approach design probes[15], which encourages participants to respond in a creative and insightful manner, providing a deeper description of their experiences and perspectives [20]. The prompts differed each day in terms of the information elicited, such as behaviours, opinions, perceptions, senses, or emotions.

We wanted to elicit data about specific journeys rather than about transportation or mobility in general. The structured diary format deliberately posed questions about specific journeys that had been undertaken, providing a more immediate and specific form of reflection on JX than would have been possible through interviews.

Questions were designed to capture the experiences and behaviours of individuals as they travel, including factors such as the mode of travel, route taken, interactions with their surroundings and our participants' experiences.

In late 2022, diary kits were distributed to participants. The diary was designed for participants to complete over 14 days. Participants were asked, where possible, to fill out the kits on a daily basis. Some participants needed more than 14 consecutive days to complete the kit, as it coincided with vacations or other responsibilities.

3.1 Designing the Diary

We aimed to gather information about the senses, activities, and emotions that were present during participants' journeys, and how these related to the environment, events, and situations that occurred during the trip.

The participants were instructed to review the diary at the start of each day before embarking on any journeys, and to be mindful of the day's requirements. We chose to use paper diaries over a digital tool in this early exploratory work because we wanted to test our ideas with minimal implementation. Writing, drawing routes on paper maps, and (for example) affixing stickers, are familiar tasks for most people. A paper diary also affords mutation and the breaking of "the rules" more easily and intuitively than digital methods and we wanted participants to be able to express themselves freely, without being overly constrained by the diary itself.

Day 1 focused on contextual factors that we thought might be important, including information about participants' primary mode of transportation, the area they live in, their preferences and dislikes, the routes they most frequently take, and their habits. On subsequent days, the diary focused on the daily journeys of the participants. It distributed the focus between activities, senses, and emotions.

The focus of the diary was on gathering information about the specifics of participants' journeys, such as the route taken, the purpose of the journey, transportation modes, participant's thoughts about the journey, activities engaged in during the journey, changes participants would make if they were to take the journey again, and their level of comfort.

Senses We included questions about which senses participants engaged during their journeys and their relationship with the routes and environment; prior research has considered visual appeal [27] and smell [32].

Emotions We included questions that focused on the affective elements of journeys; for example, asking how the journey had

triggered particular emotions. Additionally, the diary included open-ended questions about the participants' perspectives, such as what surprised them or what was negative or positive about the journey that day. We hypothesised that emotions would play a crucial role in comprehending JX.

Structured Journey Report We included some highly-structured "journey reports" that used a fill-the-blank format. We wanted to compare how short standardised answers would compare to more flexible, but potentially challenging or intimidating, free-text questions.

Hypotheticals Some questions asked participants to engage in hypotheticals, for example, "*what would you change if you did this journey again?*" We expected this question, for example, could reveal things that negatively impact JX.

Maps & Stickers On 11 days, participants were asked to complete maps with route information and respective themes for each day such as activities, senses and emotions. In a future system, we anticipate that the actual collection of a route could be automated through location sensing, but in our low-fidelity diary drawing on a map was a practical way to collect route data, which could then be annotated with additional information using sticky notes and stickers. Maps were personalised for each participant so that they showed the specific areas where each participant lived and frequently travelled (based on information collected at recruitment). The stickers were designed as a way for participants to express their potential actions, senses, and emotions. We provided stickers for concepts that we felt would be useful, plus blank stickers in case participants wanted to create their own. These tools required less writing and offered a different way for participants to express themselves; we expected that maps would be an intuitive way for people to express the links between their experiences and the spatial aspects of the journey.

Photographs Participants were asked to take photographs based on the activities for each day, such as taking a picture of where they feel safe in the street, the best place to walk, or of things that represent parts of their journeys. These questions focused on what the participants were doing, the areas they were passing through, the objects or environment they were interacting with, and the places that evoked certain emotions or sensations for them. Photographs have been used to understand mobility, such as in Desjardins et al.'s photo-journeys [11], and using a combination of personalised diary content and photographs has been considered a powerful approach to capture participant experiences [21].

Historical & Idealised Experiences On 3 days, participants were asked open-ended questions to explore their ideal trips, worst experiences, opinions, likes and dislikes. The goal was to provide a space for them to share their stories and thoughts and to give us a broader understanding of notable journey experiences that they'd had in the past; we expected that extreme or unusual experiences were unlikely to occur during the 14-day study period. Participants were given the freedom to answer the questions in their preferred method, whether it be through drawing, writing, or creating a collage. They were not restricted in how they answered these questions.

3.2 Participants

Participants for the study were selected using an online survey advertised through various channels at the University of Southampton. The survey included questions about their demographics and travel patterns. The study was completed by 11 participants, 7 women and 4 men, with diverse ages; with 2 aged 20-30, 3 aged 30-40, 2 aged 40-50, 3 aged 50-60, and 1 aged above 60 years old. Five participants resided within the city limits, while the remaining six participants lived outside the city or in other nearby cities. The study involved our institution's students, staff, and retirees. 11 out of 16 diary kits were returned, resulting in 1568 pages of the diary, 164 pages of maps, and 105 photos collected. Five participants dropped out due to obligations or vacation, one withdrew, and the rest were deemed withdrawn after 2 months of no response to emails.

3.3 Analysis

We took a "Big-Q" approach to analyse the collected data and conducted a thematic analysis [3] of the returned kits. Our overall aim is to understand more about how our participants think about their journeys and the role that our kits played in the reflective data that we received. We adopted a phenomenological approach to comprehend the data, as we were interested in understanding JX and recognized that people's journey experiences must be understood as subjective, lived experiences. However, we also hold a constructionist perspective on how the diary kits played a role in the indirect communication we had with the participants. The concepts integrated into the diary kits and the social context of the study are significant elements that influence how our participants interpret and perceive their journeys.

Analysing the diverse data formats, including geospatial data, photos, stickers, drawings, collages, and written descriptions posed a challenge. In order to begin our analysis, we familiarised ourselves with the dataset and compiled a Miro board with scanned copies of the returned kits, annotated maps, and participants' photographs. We used Miro's "sticky note" feature to apply codes.

However, we found that analysing the data using a purely "bottom-up" inductive approach was challenging due to the sheer range of factors and the relative richness of the mixed-media data elicited by the kits. Consequently, we adopted a more deductive approach in our second pass of the data, classifying each response under six a-priori "journey layers" - aspects of a journey that we had identified as potentially important to JX during the first coding and literature review. Those layers are described in Table 1. The layers provided a high-level way to think about how the different prompts had elicited information from participants, but we deliberately kept an open mind to the suitability of the layers for classifying information, taking a more inductive approach to think about data that seemed to fit "within" the layers (i.e. sub-codes) or "between" the layers (i.e. those that were difficult to classify using the layer concepts).

4 INITIAL RESULTS

We used our a-priori themes to deductively classify the collected data. We found that the diaries contained responses related to each of the themes, and we created a table that included examples of ways that the theme could positively or negatively contribute to JX. We also noted the prompts or activities in the diary which seemed to

Table 1: A-priori concepts used during the thematic analysis.

Layer	Description
Route	The series of points in space that made up a journey, typically conveyed through the maps.
Environment	Aspects of the physical environment in which the journey took place.
Place	A place is a location that has acquired some social significance, either widely or to a specific person (e.g. [7]).
Logistics	The logistics of the transport system, which include physical infrastructure like footpaths and bus stops, but also more abstract concepts like bus routes.
Activities	The activities that take place during the journey such as grabbing a coffee or picking up children.
Purpose	The motives behind a journey, the reason or reasons that it is undertaken.
Reaction	The emotions, reflections and thoughts that a journey prompts for somebody.

be most closely associated with the theme. Table 2 in the appendix shows a summary of each theme. Initially, we obtained favourable evidence supporting the existence of our predetermined themes as factors affecting JX.

4.0.1 Environment of Logistics. During analysis, we were forced to clarify our conceptualisations of Environment and Logistics. For example, should we consider "traffic congestion" as an environmental factor, or a logistical one? Through reflection on our themes, and their purpose in explaining JX, we settled on a distinction grounded in the role that a given factor plays within a specific journey. For example, a road is part of the built environment, and traffic congestion seems to be composed of a series of objects (vehicles) within the environment. However, the relevance of a congested road is quite different depending on travellers' relationship to it. In some cases, a road is a source of noise or air pollution for people walking through a park nearby; it is an environmental factor for those travellers. For travellers who are using the same congested road - for example, motorists or cyclists - the road is more relevant to their JX as infrastructure, and hence its impact on the logistics of the journey is primary. The significance of the road itself is changed by its role in affording progression of the journey and congestion becomes a frustration to the journey - a Logistical concern - rather than just a source of pollution - Environmental.

4.1 Inductive Themes

We identified additional themes to explain ideas and concepts that were not fully covered by the a-priori themes. Two noteworthy themes are *Modal Mediation* and *Progress*.

4.1.1 Modal Mediation. We observed that in some cases, the mode of transport acts as a mediator of the relationship between the individual, the wider environment, and their journey. Modal mediation, therefore, refers to the ways in which a given transport mode mediates the experience of being in a particular space. Some mediation is physical; a car or bus provides a micro-climate that is

experientially different to the outside. Additionally, different modes of transportation afford different activities; for instance, listening to music while cycling is not only ill-advised but potentially illegal.

P2 felt that walking provided a unique relationship to the social environment around her: *"I like walking along the road (main road) with lots of vegetables. I hope that someone might see me and get inspired to get an allotment,"* and felt that it would be nice to see *"people interacting and going about their daily business would be nice and feel more like a community. You can witness this in places where walking and cycling is more common."*

Other examples are less surprising: P3 mentioned that his "worst journey" occurred because the bus did not arrive when it was raining; the bus would have changed his relationship with the environment in a positive way, and is something he had sought out, unsuccessfully.

4.1.2 Progress. We interpreted from some responses that having a sense of progress during a journey is important, and seems to include both concerns that intersect with the Purpose of the journey – will I arrive on time? – but also frustration when progress is not as fast or direct as participants might want. The relationship with the space takes on an additional dimension, and it is experienced through the lens of how effectively it affords progress.

Failure to make progress is often caused by infrastructure failures, or when congestion or roadworks cause unexpected route changes. For example, P10 explained *"My worst journey was travelling home from work on the train. It is usually a long trip anyway, but when there is a disruption it can be a lot longer. One particular day, I had walked to [the local] train station, took the train to [a larger interchange]. Changed trains and went to [another nearby city]... After waiting two hours, the train company sent me to [my destination] in a taxi. So arrived home after 4 hours of travelling."* They also noted for another journey *"somewhat impatient, as travelling down country lanes – slow cars"* in response to a question about how their attitude changed during a journey.

5 DISCUSSION

Our analysis so far has centred on the thematic layers that make up JX, which we propose consists of the seven a-priori themes, plus an additional inductive theme of *Progress*.

We have drawn a distinction between journeying through a space, and being in a space; and earlier suggested a theoretical difference between the two was one reason to consider JX data as being distinct from data about the general spatial or environmental state of a city. Our two inductive themes, *Progress* and *Modal Mediation* support that distinction. They are concepts specific to JX, and which would not, conceptually, apply to analyses of being in a space where there is no transport mode to mediate the relationship and where a sense of progress is not relevant. Together, the apparent relevance of these themes to JX suggests that one of our hypotheses, that the experience of journeying through a place is different to the experience of being in a place, is supported, and therefore that there is some basis for treating JX as conceptually distinct and more nuanced than merely being in a series of consecutive places.

Based on our conceptualisation of the Purpose and Activity themes, we also observed a distinction between different types of journeys. While intuitive definitions of a journey perhaps involve

moving from an origin to a distinct destination, our participants' descriptions of their daily journeys revealed that this is not always the case. In some cases, the journey was not about reaching a destination but rather exploring a local area, getting exercise, or getting outdoors for fresh air. They were described as 'home to home' journeys by some participants, and are shaped not by the need to move from one place to another, but by the need to move.

In our initial dataset, we found it important to distinguish between linear and circular journeys as it greatly affects the relationship between the Journey and the Activity it is part of. Linear journeys are given meaning by a broader Activity that motivates them, such as work or hobbies. We draw here on the concept of Activity in the sense that it is used in Activity Theory (e.g. [24]), an overarching pattern of behaviour that is formed in response to an objectified motive. Linear journeys are shaped and motivated by the significance of the places that they link together and the actions that take place once there. The experience of the journey, while perhaps influential over modal choices, is secondary to the need to move in service of the overall Activity. In contrast, circular journeys are more often motivated by the experience itself such as walking for leisure, enjoying sights or wellbeing. Undertaking the journey is not subordinate to a higher-level activity, it is the Activity – a way of meeting needs like relaxation or socialisation. Circular journeys are conceptually similar to journeys made for tourism purposes (e.g. [16, 19]) and because of the conceptually distinct way in which they relate to Activity, we suggest that JX considerations are likely to be different between the two categories.

6 CONCLUSION AND FUTURE WORK

In this paper, we present our initial explorations of journey experience (JX) data collection using paper-based diary kits. Our initial interpretation of the data shows that JX is mediated by transportation modes, as well as trip purpose and constraints, and hence that understanding and modelling JX requires a focus on how journeys create an experience that is different to being in a space. We propose a set of seven layers, based on a-priori theory and inductive analysis, which can form the basis of further exploration of JX data collection.

In the future, we plan to expand our participant pool and translating our data collection method into a digital tool. We plan to combine automatic sensing (such as route, time, day, and environmental factors) with human-generated experiential data to minimise the human effort required for data collection and obtain valuable insights into participants' experiences. We would like to further explore the distinction between linear and circular journeys, and how they impact JX and JX data collection. Additionally, we will further explore different methods of elicitation to collect insightful data and formalise our analysis based on the insights gained from this study. Finally, we will develop a more structured approach to analyze the larger data set.

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7 APPENDIX 1: SUMMARY OF THEME MAPS

Positive JX	Negative JX	Related Data Sources	Observations
<i>Route</i>			
Short routes, passing by pleasant areas	Being forced to change their route, walking adjacent to main roads, crossing roads or junctions, finding another gas station, traffic jams	maps, structured journey report, hypotheticals, factors of negative experience	Desire for real-time traffic information, having other route options, more efficient bus route options, having paths for walking that avoid busy roads and crossings
<i>Environment</i>			
Pleasant smells, silent streets, physical features such as green spaces, points of interest (POI), amenities like bins or seats, good weather, animals, fresh air	Disruptions such as constructions or accidents, unpleasant smells such as exhaust fumes, traffic noise, crowd, traffic, other road users, rain	maps, structured journey report, senses	The environment plays a role in determining the participants' choice of transportation, particularly in regards to the temporal factors such as the number of people on the bus, the weather, or the time of day.
<i>Places</i>			
favourite places, meeting points with friends, historical significance, work, home, safe areas, activity areas	main roads	maps, structured journey report (transport mode), positive/negative surprise, photos	Perceptions of transitional places, like main roads or the nearest station, were affected by factors such as the physical environment and the mode of transportation used.
<i>Logistics</i>			
available transport modes, nearby bus stops, preferred transport mode being available for a journey, short travel time, direct routes	route/mode transfers, confusion over routes, lack of service coverage, long travel time, lack of infrastructure and facilities, uneven ground and terrain	maps, structured journey report (transport mode), comfort, positive or negative preferences, photos	Desire for more efficient bus routes, closer distances to public transport, less roadworks
<i>Activities</i>			
grabbing a coffee, running into friends or neighbours, picking up colleagues, stopping by a shop and gas station, taking a photo, listening music, sitting in the park	waking up late, arriving home late	maps, structured journey report, idealised experiences	Desire for increased social activity and interaction, different modes afford different activity during a journey
<i>Purpose</i>			
wellbeing, exercise, meal with friends, going to stable, going to shop	daily commute to work, arriving later than required	maps, structured journey report, experience, positives/negatives	Participants' perception of their JX was often linked to their transport mode, rather than their journey's purpose.
<i>Reactions</i>			
serenity, apprehension, contempt, surprise, humour, joy	pain, fatigue, feeling sweaty, feeling rushed, annoyance, vigilance, anticipation of negative experiences	maps + stickers, experience, positives/negatives, idealised experiences, photos	photos taken by participants were often challenging to understand in terms of how they related to their overall journey experience, but helped to illustrate this theme in particular.

Table 2: Thematic map of a-priori themes

Positive JX	Negative JX	Related Data Sources	Observations
<i>Progress</i>			
Proceeding on schedule, getting closer to a destination	Falling behind schedule, having to take geographically indirect routes, periods spent not moving	maps, hypotheticals, structured journey report	Progress has temporal and geographic elements. Participants want to feel that they are getting closer to a destination, and/or executing against a plan

Table 3: Thematic map of inductive Progress theme